

ECMWF's societal impact through service provision, partnerships and collaborations

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ABSTRACT

The European Centre for Medium-Range Weather Forecasts (ECMWF) mission is to deliver high-quality global medium-range numerical weather predictions and monitoring of the Earth system to its Member States, but its benefits reach far beyond thanks to the vast network of relationships that ECMWF has developed since it was created. There are multiple reasons for the success of ECMWF: scientific and technical excellence, high-quality service provision, its intergovernmental status, and an appropriate level of investment and support from its Member States. This paper concentrates on discussing how partnerships, collaborations and interaction with the user community contribute to shaping ECMWF activities and to maximizing its societal impact.

1. Introduction

The primary purpose of the European Centre for Medium-Range Weather Forecasts (ECMWF) since its establishment as an independent intergovernmental organisation in 1975 is to pool Europe's meteorological resources to produce weather forecasts for its Member and Co-operating States. Currently, 23 countries joined ECMWF as Member States and 12 as Co-operating States. The development of the new strategy for 2025–2034 has allowed ECMWF to reflect on its impact through a study on the societal benefits of ECMWF commissioned to Citizing, a French consultancy company. The primary purpose of the study was not to monetize the socio-economic benefits, but rather to provide an overview of the breadth of activities carried out by ECMWF in collaboration with the wider meteorological community (Fig. 1) and capture their societal impact via numerous case studies. This paper is primarily based on the outcome of the study. By publishing in the open literature, the results become widely accessible and hopefully useful in the broader context, helping to justify investment in the science, computing and other infrastructures for accurate and authoritative weather and climate public services in Europe and other parts of the world.

ECMWF works in collaboration with the National Meteorological and Hydrological Services (NMHSs) and research institutions of its Member and Co-operating States to develop modelling capabilities, design new products and evaluate and diagnose forecast quality. With

headquarters in Reading (UK) and sites in Bologna and Bonn, the organisation currently employs around 520 staff from 35 countries. ECMWF has been producing operational medium-range weather forecasts since its origins and in time has developed a capability for ensemble prediction extended to sub-seasonal and seasonal time ranges and a broad portfolio of earth system monitoring and prediction products in the domains of atmospheric composition, climate, ocean, ocean waves, sea ice, hydrology, fire and land surface. ECMWF is also a major user of satellite data, having co-operation agreements with the European organisation for the Exploitation of Meteorological Satellites (EUMETSAT) and the European Space Agency (ESA), and of many other observations including weather stations, aircraft, ships, and buoys. Altogether, ECMWF operationally processes more than 100 million observations daily.

The peculiarity of ECMWF is in its dual nature, being an inter-governmental organisation that is simultaneously a research institute and a 24/7 operational service, producing and disseminating numerical weather predictions (NWP) and monitoring of the Earth system primarily for the official duty of the NMHSs of its Member and Co-operating States, but with an extensive catalogue of products available to other partners and users. To perform its tasks, ECMWF operates one of the largest high-performance computing facilities (HPCF) in Europe, connected to the NMHSs of the Member and Co-operating States through high-speed telecommunication and digital links. Data

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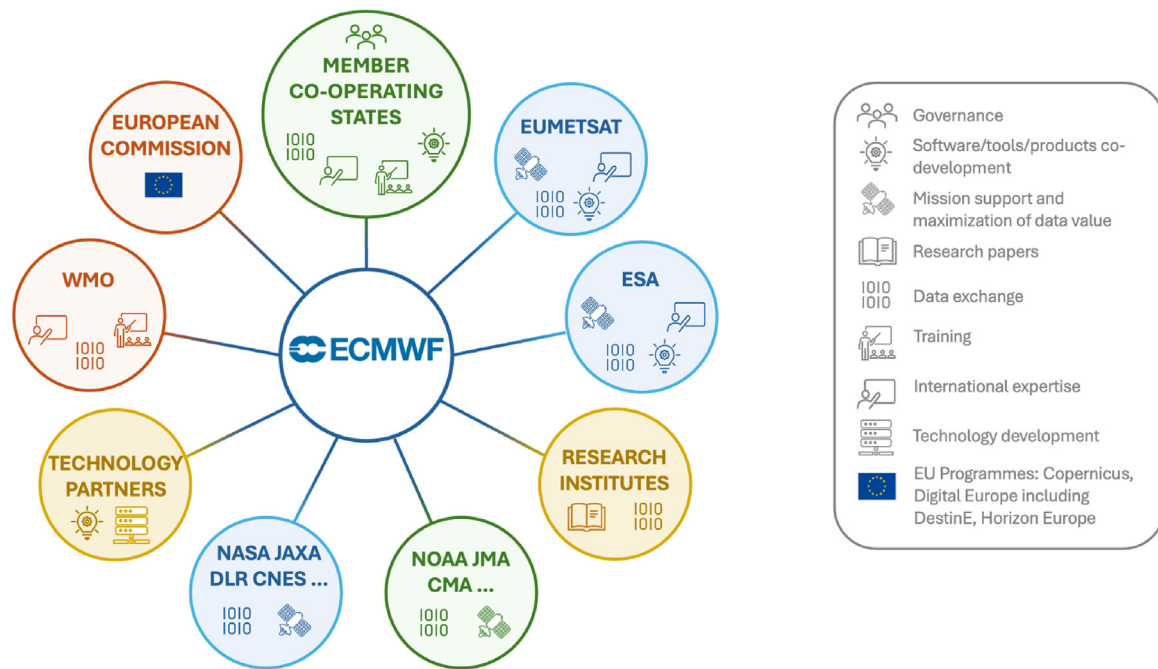


Fig. 1. Schematic representation of some of ECMWF strategic partnerships as of 2025 and discussed in this paper. The Member and Co-operating States are the main partners, collectively participating in the governance of ECMWF and defining the strategic priorities. The main activities considered in the paper include enhancing the use of satellite observations, research in data assimilation and modelling, development and provision of software, tools and products, technology development, training, activities conducted on behalf of the European Union in support of the Copernicus and Digital Europe programmes, and the provision of scientific and technical expertise in support of international initiatives.

ingested and produced by ECMWF are regularly archived and constitute today the world's largest meteorological archive, containing more than 1 Exabyte of data.

Since its creation, ECMWF's influence has grown significantly thanks to the support of its Member and Co-operating States and the increasing number of scientific partnerships and collaborative networks worldwide built with the NMHS, the World Meteorological Organisation (WMO), space agencies, the European Commission, research centres, universities, and the private sector. ECMWF is the entrusted entity to operate two of the six services of the European Union's Copernicus Programme, specifically the Copernicus Climate Change Service (C3S) and Copernicus Atmosphere Monitoring Service (CAMS), for the production and dissemination of environmental monitoring and forecast data and information. ECMWF also runs the flood and forest fire forecasting systems operationally on behalf of the Copernicus Emergency Management System (CEMS), which is managed by the Joint Research Centre (JRC). Over time, the meteorological landscape slowly evolved, but in recent years, the signs of substantial changes ahead have manifested even more clearly. The political and societal relevance of weather forecasting and environmental monitoring products and services (both via traditional channels such as governmental agencies and innovative private enterprises) and their use to underpin policymaking have dramatically increased. HPC and cloud technologies are also rapidly evolving, and their uptake in new domains may influence their future development. International initiatives, such as the European Union's Destination Earth, are promoted to enable digital environmental programmes. Finally, the traditional physics-based weather forecasting value chain is being disrupted due to the development of Machine Learning (ML) models.

2. The value of weather and environmental monitoring and forecasting information

Weather forecasts positively impact individuals, businesses and entire countries not only by providing key information for the protection of life and property but also by facilitating innovation, knowledge

creation and collaborative work across all societal sectors and internationally. Improvements to operational NWP models are proven to provide significant socio-economic gains by making better information available and, therefore, supporting decision making. Accurate weather forecasts enable businesses to strategically plan production, consumption and distribution of resources to minimize risks and maximize value creation. Studies show that improving the accuracy of weather forecasts has a significant positive impact on the output levels of all industries — between 1.5% and 3% for every 1% increase in weather forecast accuracy, with sectors more dependent on weather forecasts experiencing a more pronounced effect (Shen et al., 2024). Furthermore, the use of historical weather data allows industries to assess their sensitivity to weather variations, enabling proactive planning and strategy adaptation and, therefore, improving business performance (Brown et al., 2019). Accurate weather forecasts are, of course, not just important for boosting businesses but primarily for safeguarding populations and property, especially in case of high-impact events. It has been estimated that in the United States alone, reducing the standard deviation of forecast errors by 50% would save 2200 lives per year (2400 lives per year accounting for the effects of climate change) by 2100 (Shrader et al., 2023). It has also been valued that giving just 24 hours' notice of an impending hazardous event can reduce socio-economic damage by 30% (Global Center on Adaptation, 2019). Hallegatte (2012) estimated that the use of early warning systems in Europe allows society to avoid between € 460 million and € 2.7 billion in lost assets per year. This represented between 0.003% and 0.017% of the European GDP at the time. The same study determined that if early warning systems were modernized in all low- and middle-income countries to the level of Europe, on average, an additional 23,000 lives could be saved per year, and between US\$300 million and US\$2 billion of annual asset losses could be avoided globally. By additionally quantifying the benefits to “well-being”, moving the analysis beyond short-term financial matters and considering long-term socio-economic factors such as poverty, public health, education and livelihood productivity, Hallegatte et al. (2017) found that globally US\$22 billion in additional benefits could be produced on average per

year by providing universal access to improved weather forecasting and early warning. In the Western Balkans alone, the World Bank conservatively estimated the potential economic effect from more specific, rapid and clearly communicated forecast and early warnings for the agriculture, transport, construction, and energy sectors to be US\$ 50–100 million per year (WMO, 2024). Furthermore, Kull et al. (2021) examined the additional socio-economic value that would be created if surface-based observations from Antarctica, Africa, South America, the Pacific and parts of Asia increased to meet the WMO Global Basic Observing Network specifications. This enhanced coverage would, in turn, improve weather forecasting quality globally and in particular in these data-sparse regions and could deliver additional global socio-economic value of over US\$5 billion annually.

Benefits are not only derived from short and medium-range forecasts. There is an increasing demand for longer lead times in weather predictions, giving more time to prepare and enabling very early actions to minimize the economic impact of extreme events. The Red Cross and Red Crescent Societies are starting to use sub-seasonal forecasts to strategize for weather disasters, for instance, to plan where to move emergency supplies when a tropical cyclone threatens to hit a region. Furthermore, Forecast-based Financing (FbF)¹ enables access to humanitarian funding for early action based on in-depth forecast information and risk analysis to anticipate disasters, reducing their impact, if possible, and human suffering and losses. The allocation of financial resources is agreed in advance, together with the specific forecast threshold that triggers the release of those resources. As for seasonal forecasts, these can allow firms to better assess long-term risks and, therefore, make better investing decisions. Lemoine and Kapnick (2024) show that financial traders price the uncertainty generated by the US National Oceanic and Atmospheric Administration (NOAA) El Niño and winter outlooks, affecting firms respectively at US\$13.4 trillion and US\$5.7 trillion, and that a 1% improvement in the skill of the El Niño outlook reduces firms' exposure to a one standard deviation shock by \$18 billion and induces traders to spend an additional \$2 million hedging the outlook's news. Improved seasonal forecasts also contribute to a better understanding of the climate dynamics which, over time, are expected to produce better climate mitigation and adaptation policies. Hope (2015) calculates the value of better information relative to one aspect of climate, i.e. the rise in global mean temperature that would be caused at the end of 70 years by a 1% per year increase in the concentration of CO₂ in the atmosphere, and shows that halving the information uncertainty has a net present value of about \$9.7 trillion (year 2005 US\$) if accomplished in time for emissions to be adjusted by 2030.

The full benefit of weather and environmental information, however, can be secured if such information is available timely and free of cost. The latter aspect is particularly important in Least Developed Countries. Research shows that open access to high-quality weather data enhances countries' ability to prepare for and rapidly alert and respond to natural disasters (Scheip and Wegmann, 2021), leading to better evacuation plans, resource allocation, and overall disaster risk reduction, thereby saving lives and reducing economic losses. Free access to weather data is also crucial in understanding the economic impacts of natural disasters: by analysing the costs associated with weather-related disasters, governments can allocate resources more efficiently and implement targeted mitigation measures (Msemo et al., 2021) and open-access weather information facilitates the identification of vulnerable populations and helps in tailoring response efforts to meet their specific needs (Burger et al., 2019). In addition, facilitated access to data through open and free licenses accelerates research by

transforming scientific findings rapidly in operational solutions: researchers can access larger and diverse datasets without the constraints of closed licensing issues, leading to more comprehensive studies and faster advancements in science and technology. The European Commission funded a study on the socio-economic impact of the Copernicus Programme in the EU (PwC for the EC, 2016), partly related to the value created by open data. The net effect is highly positive with a conservative economic (GDP) result of €38.3 million in 2015 and an *ex-ante* cumulative enabled revenues of €445.1 million between 2016 and 2020.

3. ECMWF value added through international collaboration

Weather and climate events impact populations and businesses locally but must be studied at a global scale. To anticipate phenomena affecting a region of the world, it is often necessary to understand the evolution of the state of the atmosphere in other regions. This encourages international collaboration to facilitate the sharing of data, codes and information, resulting in the reduction of forecasts' cost production, improved forecasting accuracy and a better understanding of weather and climate patterns. ECMWF has strategically pursued a wide range of diverse partnerships and relationships with the broad meteorological and climate community including the NMHSs of Member and Co-operating States, international organisations (WMO, EUMETSAT, ESA), the European Commission, governmental agencies and commercial companies worldwide, that can take many forms such as collaboration on research programmes, strategic orientations and participation on advisory boards, improvements of NWP, co-development of products and applications, monitoring of the quality of observations, training and support in the integration of ECMWF's output in products and services targeted to many societal sectors and the public at large. It is through such entities that most of ECMWF's socio-economic impact is manifested.

4. Research work

Research collaborations bring together scientists from different disciplines, backgrounds and institutions, fostering the exchange of diverse perspectives, ideas, resources, and expertise. This diversity can spark innovative approaches and creative solutions to complex problems. Studies have shown that collaborative innovation activities, capabilities, and knowledge sharing significantly influence innovation performance in the public sector (Widyasari and Nasution, 2022) and can lead to the development of new applications allowing for more robust and reliable results (Wagner et al., 2015; Shao et al., 2020). Since ECMWF activities are linked to many aspects of global environmental monitoring and prediction, research is conducted through collaboration with most members of this research ecosystem.

To improve NWP, it is paramount to invest in research to guide the design of the Earth observations network and enhance the use of observations. In this sense, ECMWF works closely with its Member States and other partners such as space agencies, primarily EUMETSAT and ESA, and with significant collaborations with NASA, NOAA, the Japan Aerospace Exploration Agency (JAXA), the China Meteorological Administration (CMA), the European Commission and the WMO. This work enables ECMWF to process and use operationally a large number of satellite instruments (Fig. 2). ECMWF and EUMETSAT jointly invest in research on advanced techniques to extract information from satellite data optimally. For instance, ECMWF's groundbreaking all-sky approach to microwave data exploitation has enabled the innovative extraction of humidity and temperature information in the presence of clouds and precipitation. It is the first of its kind used in any operational weather model worldwide. This work has benefited from EUMETSAT support through its fellowship program and the NWP Satellite Application Facility collaboration (including the UK Met Office, Météo-France and Germany's DWD) and ESA's work on the development of satellite

¹ A programme that enables access to humanitarian funding for early action based on in-depth forecast information and risk analysis. The goal of FbF is to anticipate disasters, prevent their impact, if possible, and reduce human suffering and losses. <https://www.forecast-based-financing.org/>.

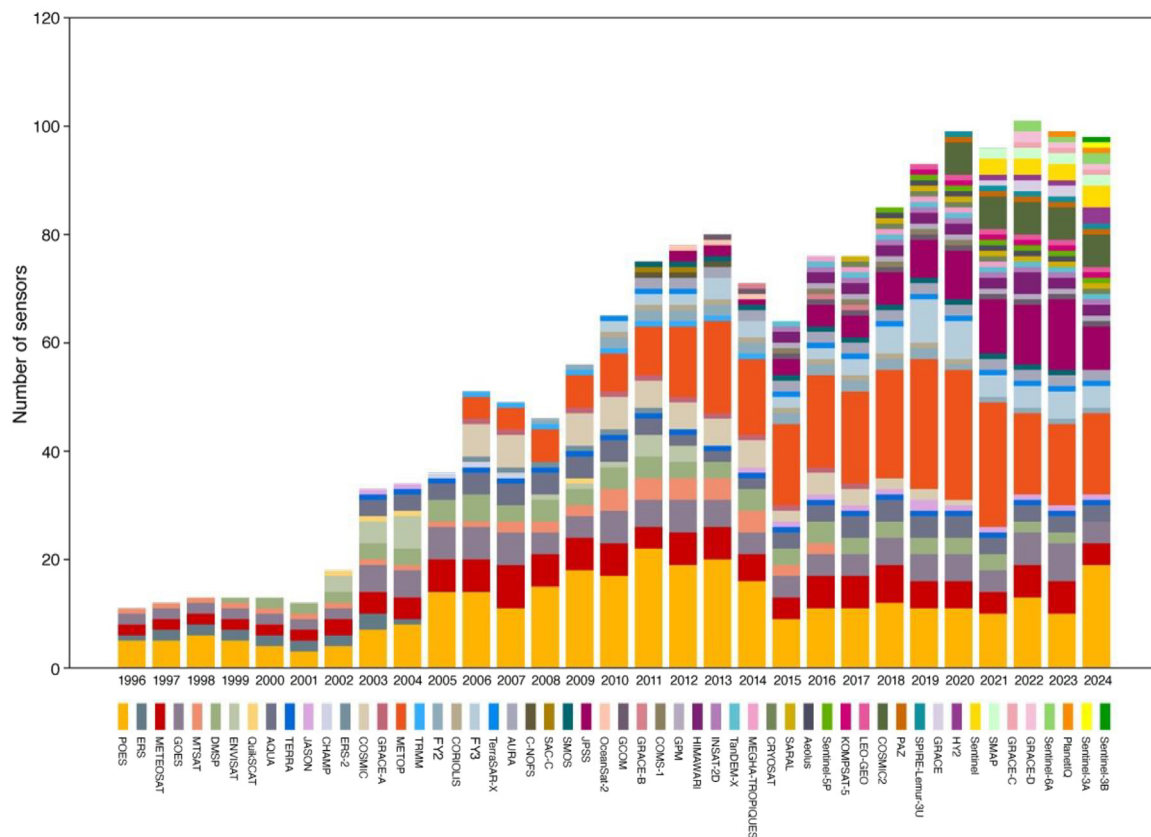


Fig. 2. ECMWF benefitted greatly from the large investments in Earth observations. The figure shows the number of satellite instruments processed and used operationally at ECMWF. Satellite instruments changed dramatically the observing system since the late 1990s. EUMETSAT Metop-A (2006) was Europe's first operational polar-orbiting satellite, with a suite of novel and advanced instruments, including the first fully operational GPS radio occultation instrument. ECMWF assimilates data from both operational and research satellites (such as ESA's Earth Explorers) quickly after launch. In more recent years, ECMWF started using data from satellites operated by private companies (e.g. SPIRE).

radar assimilation for the forthcoming EarthCARE satellite mission. The collaboration between ECMWF, EUMETSAT and ESA also extends to the Copernicus programme, in particular in CAMS and C3S.

ECMWF provides expertise relevant to the development of ESA's Earth Observation programs through participation in mission advisory teams and mission selection and planning activities. For instance, ECMWF participated in the Aeolus Data Innovation and Science Cluster, an international expert consortium to study and improve the data quality of the Aeolus satellites derived products. The mission's objective was to demonstrate that observations from a doppler wind lidar measuring wind profiles throughout the atmosphere can improve the weather forecast. Many agencies contributed to the mission, including the German Aerospace Center (DLR), French Centre National de la Recherche Scientifique, Royal Netherlands Meteorological Institute (KNMI), Leibniz Institute for Tropospheric Research (TROPOS), Météo-France, the software company DoRIT, and NASA. ECMWF's role was to validate and calibrate the data, optimize their utilization, and assimilate and measure the effect on the quality of weather forecasts.

In a different context, but with the same general aim to improve the observing network, the WMO commissioned to ECMWF a study to estimate the impact on the accuracy of the forecast of various configurations of the Global Basic Observing System to guide investments made by the Systematic Observations Financing Facility (SOFF) to help NMHSs particularly in Least Developed Countries and Small Island Developing States.

One of the main objectives of ECMWF's research activities is the continuous improvement of the Integrated Forecasting System (IFS) model and related elements of NWP. A prime example is the research collaboration with Météo-France in developing IFS/ARPEGE. Since about 30 years ago, both ECMWF and Météo-France have run

daily a global high-resolution forecast model, IFS at ECMWF and ARPEGE (Action de Recherche Petite Echelle Grande Echelle) at Météo-France (Pailleux et al., 2015). This collaborative project started in 1987 and led to the operational implementation of the models in 1992 at Météo-France and in 1994 at ECMWF, with regular upgrades ever since. Important outcomes of this collaboration are the development of 4D-Var data assimilation to combine model forecasts with observational data (Rabier et al., 2000; Thépaut and Courtier, 1991) and the Object-Oriented Prediction System (OOPS), a framework for running 4D-Var. The implementation of 4D-Var in 1997 was a first worldwide and became a method of choice elsewhere as it was subsequently adopted by several global NWP centres. This advanced data assimilation technique led to a much more consistent way of using observations distributed in time and a larger impact of satellite observations on the forecast skill. The effect of the combination of this groundbreaking technique and the advent of more advanced satellites led to a clear performance jump in the ECMWF forecasts, as seen in Fig. 3. More recently, the changes introduced to the ensemble forecast in 2023 that improved tropical cyclone tracks and intensity forecasts are in part a result of the IFS/ARPEGE project (ECMWF, 2024b). Collaboration on IFS/ARPEGE is also crucial for adapting codes to accelerated architectures based on Graphics Processing Units (GPU). This software is also co-developed with the ACCORD consortium, the largest international weather prediction research consortium, comprising 26 European and North African NMHSs. The ACCORD consortium aims to enable participating NMHSs to deliver improved forecasts, particularly for extreme weather events, by developing high-resolution (1 km or less) NWP for short-range (1–3 days ahead) forecasts at the regional/local level.

By combining diverse expertise, resources, and perspectives, collaborative efforts in research can lead to innovative solutions. For instance,

the Nucleus for European Modelling of the Ocean (NEMOVAR) data assimilation system is a variational scheme which has been developed specifically for use in the NEMO ocean model (Maded and the NEMO System Team, 2023), developed jointly by the UK Met Office, ECMWF, Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique (CERFACS), and Institut National de Recherche en Informatique et Automatique/ Laboratoire Jean Kuntzmann (INRIA/LJK). The system can assimilate in-situ temperature and salinity profiles, sea surface temperature, sea level anomaly and sea-ice concentration data. NEMOVAR improves the skill of seasonal forecasts and the mean state and interannual variability of the ocean estimation in most regions, including the Equatorial Atlantic, a traditionally challenging area (Mogensen et al., 2012). Other prime examples of collaboration to tackle long-term and difficult research challenges in global Earth system prediction are with AFESP (Research Programme on Advancing the Frontiers of Earth System Prediction), together with the University of Reading, the UK Met Office and the National Centre for Atmospheric Science, and with CESOC (Center for Earth System Observation and Computational Analysis, comprising the University of Bonn, the University of Cologne, and Forschungszentrum Jülich) and DWD on physical parametrization and visible radiances.

While physics-based NWP models are the backbone of current forecasting, ECMWF acknowledges the increasing potential of ML techniques to enhance the process. ML techniques are being investigated in data assimilation to enhance the accuracy of predictions. Several studies have shown that ML weather forecasting tools, such as Huawei's Pangu Weather, NVIDIA's FourCastNet and Google DeepMind's GraphCast, can predict weather with accuracy and faster than traditional NWP models in medium-range, short-range and ensemble forecasts (Bi et al., 2023; Charlton-Perez et al., 2024; Ben Bouallègue et al., 2024). ECMWF entered the arena with the real-time implementation of its data-driven Artificial Intelligence Forecasting System (wrf). The AIFS is underpinned by a toolkit, called Anemoi, which provides high-performance building blocks and pipelines to create and train massively parallel AI-driven forecast models. AIFS produces highly skilled forecasts for upper-air variables, surface weather parameters and tropical cyclone tracks (Lang et al., 2024). ECMWF runs in real-time AIFS and other open-source AI models mentioned above to compare their performances and to document their strengths and weaknesses. ECMWF is now also developing an ensemble version of AIFS. These developments are also being used as a basis for collaborations with the NMHSs of ECMWF's Member States. In particular, envisaging a wider use of Anemoi, ECMWF is funding a project led by MET Norway and MeteoSwiss, with the participation of (currently) other 12 NMHSs across Europe to build on Anemoi and create data-driven weather forecasting models at regional or global scale, allowing the NMHSs to bring datasets and ideas for the architecture configuration, and train their own models for different applications (ECMWF, 2024a). It is important to note that ML developments in weather and climate forecasting worldwide rely heavily on the use of reanalysis data as learning datasets. AIFS, Pangu Weather, FourCastNet and GraphCast, as well as other ML models not mentioned here, use the ERA5 reanalysis, produced by ECMWF and funded by the Copernicus programme. ECMWF is already developing the next generation of reanalyses (ERA6), which will be available at higher resolution. Since future developments of ML techniques largely depend on improvements of reanalysis products, it is reasonable to expect that ERA6 will contribute to increasing the speed and depth with which ML will impact the meteorological sector.

Although for ECMWF the number of scientific and technical publications does not necessarily represent a key performance indicator in the same way that it does for purely academic institutions, it is still relevant to note that in the last twelve months, ECMWF staff published 140 peer-reviewed articles and, between 2020 and 2024, 48 book chapters were published, as well as 54 technical memoranda, 106 reports, 18 newsletters and 65 newsletter articles.

5. Data, forecasts and services

ECMWF is a world leader in global medium-range weather forecasting. ECMWF's ensemble forecast (or ENS) is an ensemble of 51 forecasts designed to indicate the range of possible weather conditions out to 15 days ahead, including the probability of the occurrence of high-impact events. According to its Technical Advisory Committee, ECMWF leads other centres on various medium-range forecast verification scores, including ensemble scores for precipitation, upper-air weather parameters, stratospheric forecasts, ocean wave parameters and spread-skill relationship. This is why the ECMWF forecasts are sometimes referred to as the "gold standard" of global weather models. ECMWF regularly updates its forecasting system, typically once a year, after thorough tests, to improve the forecast skill performance. In 2023, for instance, ECMWF improved the spatial resolution of ENS from 18 to 9 km, leading to forecast improvements between 2% and 6% for surface variables and between 1% and 3% for upper-air variables (ECMWF, 2024b).

ECMWF's medium-range forecasts are extremely important for the NMHSs of its Member and Co-operating States. All of them use ECMWF's data in the medium range as their primary products for their weather forecasting services if they do not produce such forecasts for capacity or strategic reasons, or as a complement to their own products and services. Many of them use ECMWF data as boundary conditions for their short-range forecasts. For instance, a number of European countries have weather-based heat health warning systems (HHWS) in place, which provide graded heat alerts. The majority of the HHWS are run by the NMHSs. Of the fourteen European countries analysed by Casanueva et al. (2019), nine used ECMWF raw or derived products, including the temperature extreme forecast index (EFI), indicating how extreme predicted temperatures are with respect to the forecast model climate. HHWS were instrumental to plan interventions ahead of the heatwave that affected Europe in 2022, which was well predicted by ECMWF medium-range forecasts (ECMWF, 2022) but still caused an estimated 61672 deaths (Ballester et al., 2023). Another example of the use of ECMWF medium-range forecasts and EFI in particular was in connection with cyclone Filomena which brought exceptional snow precipitation to the interior of Spain, including Madrid, in 2021 (ECMWF, 2021). The precipitation was followed by an extreme cold wave affecting the whole peninsula. Red warnings were set up by the Spanish NMHS AEMET in most of the central and northeast quadrant of the peninsula due to the high accumulated precipitation expected in ECMWF's ensemble forecast for several consecutive model runs.

Increasingly, also NMHSs worldwide are using ECMWF data to initialize their limited area models or to use them directly as forecasts in their services. NOAA makes ECMWF forecasts available to the National Hurricane Centre and includes them in the National Blend of Models (NOAA, 2024). In 2023 alone, thirteen NMHSs worldwide subscribed to licences to get ECMWF forecasts in real-time. Many more NMHS (currently 37, but others are being arranged) of Least Developed Countries and Small Island Developing States have been granted access to a special dataset designed to feed initial conditions to limited area models thanks to an initiative promoted by ECMWF in support of SOFF to encourage these countries to share observations that improve the quality of global forecasts, which in turn improve the local forecasts, thus creating a virtuous circle that eventually benefits all.

ECMWF forecasts in the medium range are also used by private weather companies and other enterprises to provide specialized services in sectors such as energy, transportation, logistics, insurance, intelligence, and sporting, as well as weather applications for public use. The number of commercial licences for the use of ECMWF data rose from 103 in 2011 to 392 in 2023. For instance, Windy is a Czech company enabling 800,000 users per day worldwide (CAMS, 2024) to access comprehensive weather forecasting services using sophisticated visualization tools via an app and website. The platform

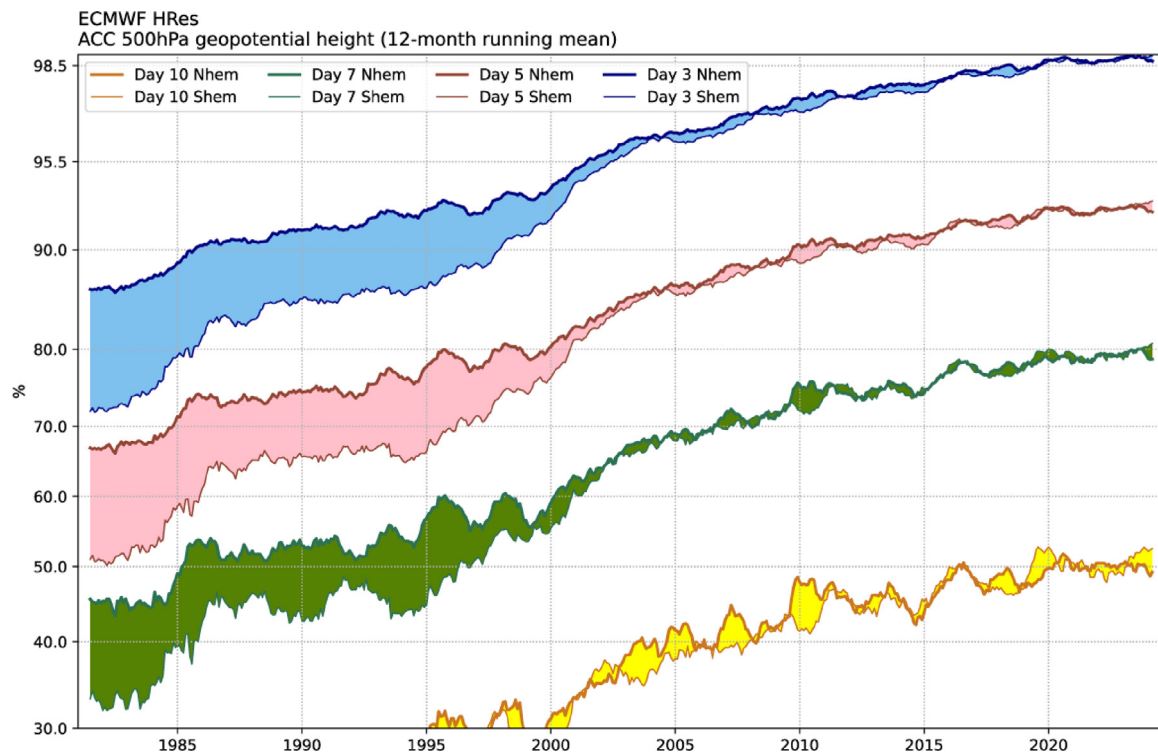


Fig. 3. Lead time of anomaly correlation coefficient (ACC) reaching multiple thresholds for the high resolution (HRES) 500 hPa geopotential height forecasts. ACC reaching 100% indicates a perfect correlation between the forecast and the verifying analysis. The figure shows the steady progress in accuracy. Note the “jump” made in early 2000s mainly attributable to the 4D-Var assimilation of satellite data, which also contributed to the reduction in disparity between the northern and southern hemispheres. In 2000 the 80% threshold was reached at day 5, while in 2020 it was reached at day 7, showing an improvement in forecast accuracy of about 1 day per decade. The vertical ordinate is linear in the Fisher transform of the ACC, $0.5(\ln(1+ACC)-\ln(1-ACC))$ (Simmons and Hollingsworth, 2002).

displays 52 layers, most sourced from the ECMWF models. Moreover, Windy services include data produced by CAMS such as air quality and ozone forecasts, and data to track the movement of aerosol plumes, from sources such as dust, volcanic eruptions, sea sprays and biomass burning. Several weather providers use ECMWF input for their services to sailors participating in the Vendée Globe, a non-stop, unassisted, single-handed round-the-world yacht race. Sailors use ECMWF data to “find the best passage and to be fast and stay safe” (ECMWF, 2021).

ECMWF also produces sub-seasonal (three to six weeks ahead) and seasonal (seven months to one year ahead) forecasts. These are important especially for the NMHSs of Member States that do not themselves produce data at these time scales. 2024 marks the 20th anniversary of sub-seasonal predictions at ECMWF. The performance of these forecasts has improved over time, especially in coincidence with the use of the reanalysis ERA5 to initialize the reforecasts, and further improvements are expected when ERA6 will be eventually adopted. As this still represents a very challenging area in capturing the Earth System’s predictability, it is not surprising that the societal impact of sub-seasonal predictions is not yet comparable to the medium range. However, applications are being tested in health (e.g. to predict the outbreak of vector-borne diseases), agriculture (to select crops and plan soil fertilization and harvest), and energy (to support hydropower management) (ECMWF, 2021). ECMWF seasonal forecasts have shown significant improvement in recent years, especially demonstrated by correctly capturing phenomena such as the transition from La Niña to El Niño in the autumn of 2022, the anomalously warm 2022–2023 winter in Europe and the warm European summer anomaly of 2023 (ECMWF, 2023b, 2024b).

Beyond weather prediction, one of the key contributions of ECMWF to society is through the Copernicus programme, by producing a range of open and free state-of-the-art environmental monitoring and forecasting data and information.

C3S provides quality controlled information about the past, present and future of the climate in Europe and worldwide. Among the most popular C3S datasets are the so-called reanalyses, describing the state of the atmosphere, land surface and ocean waves and providing the best possible picture of past weather and climate conditions. C3S reanalyses include the above-mentioned global ERA5, the higher-resolution ERA5-Land, and regional reanalysis for Europe and for two subdomains of the Arctic. Data and tools are provided for use by governments, public authorities, policy makers and private entities (Buontempo et al., 2022) across multiple disciplines. For instance, reanalyses form the basis for monthly C3S climate bulletins, and it is used in the assessment of the State of the Climate in Europe which is published annually by C3S and the WMO. ERA5 is regularly cited and used in the IPCC Assessment Reports to underpin their findings and support informed recommendations. To ensure the quality of its information, C3S invests significantly in quality management frameworks for its climate datasets, aiming to establish trust between data providers and users (Lacagnina et al., 2022).

CAMS is the leading European service monitoring air quality, climate patterns, and UV radiation levels. CAMS provides twice-daily information on global and regional air quality, inventory-based emissions, observation-based surface fluxes of greenhouse gases and from biomass burning, solar energy, ozone and UV radiation. Tens of thousands of users access CAMS platform directly and well over 200 million end users worldwide via smartphone applications, websites, or TV bulletins in Europe and worldwide (Peuch et al., 2022). One of the main features of CAMS resides in its capability to track variations in air quality either due to natural phenomena such as weather patterns or to changes in emissions. During the first phase of the COVID pandemic, CAMS observed decreases of 20% to 30% in fine particle matter levels PM_{2.5} for February 2020 compared to previous years (Chigbu, 2020) and between 20% and 40% in nitrogen dioxide (NO₂) in January–April 2020 compared to the same period in 2019 (Bauwens et al.,

2020), and could differentiate what was due to decreased human activity linked to COVID from what was linked to natural variability. Through its capabilities in source apportionment calculations, CAMS enables the daily evaluation of air quality in European cities (Thunis et al., 2021). The introduction of new components, such as monitoring anthropogenic CO₂ emissions, demonstrates CAMS evolution towards providing comprehensive environmental data for informed decision making at a global scale (Peuch et al., 2022). To support EU Member States in integrating CAMS products into their innovative air quality data and services, ECMWF currently runs the National Collaboration Programme (NCP) with 15 countries, while 5 more are being arranged.

The Copernicus Emergency Management Services (CEMS) led by the Joint Research Centre (JRC) provides early warnings of natural disasters. ECMWF contributes to the European Flood Awareness Systems (EFAS), the Global Flood Awareness Systems (GloFAS) and the European Forest Fire Information System (EFFIS). ECMWF is responsible for the operation of the CEMS hydrological probabilistic forecasting system. EFAS and GloFAS provide complementary flood forecast information to relevant stakeholders for national, regional, and global flood risk management. Use cases of GloFAS include supporting the humanitarian response to the devastating floods that affected large parts of Mozambique, Malawi, and Zimbabwe in the wake of tropical cyclone Idai in March 2019 (Harrigan et al., 2023) following a request from the Department for International Development of the UK government. GloFAS is also a main contributor to the SLAPIS early warning system covering the Niger section of the Sirba river and will be used to implement the SLAPIS Sahel project, a joint early warning system between Niger and Burkina Faso expected to benefit nearly 2 million people most vulnerable to the risk of floods in the region (CNR, 2024). ECMWF's ability to contribute to flood forecasting in the Least Developed Countries, where in situ observations are sparse, is due to the implementation of an Earth system modelling approach and the assimilation of a wide variety of satellite data.

EFFIS is an essential tool in monitoring and managing wildfires in Europe and across the Middle East and North Africa (CEMS, 2024). As part of CEMS, EFFIS offers a decision support system that aids in monitoring forest fires at a trans-national scale, harmonizing information across European and Mediterranean countries, and therefore facilitating international collaboration on forest fire prevention and fighting, particularly during trans-boundary fire events (Rodríguez-Aseretto et al., 2013). EFFIS assesses fire danger levels using the Fire Weather Index (FWI) to evaluate fire risk and provides adjusted thresholds for identifying different severity levels of wildfires, helping authorities adjust the level of response efforts correspondingly. After wildfires have occurred, EFFIS allows for the characterization of burnt areas, providing valuable insights for effective post forest fire management and appropriate reforestation efforts (Fonte et al., 2021).

The production and supply of weather and environmental monitoring and forecasts are clearly key activities of ECMWF but, even if quality is generally considered to be high, this would not in itself be sufficient to guarantee a widespread uptake, as the data require a high level of expertise to be used appropriately and interpreted. For this reason, ECMWF has put in place an extensive education and training programme. ECMWF provides training through a combination of in-person courses, online resources (material, documentation and e-learning), collaborative initiatives, and extensive support for its users. Training is also sometimes co-organized with WMO, EUMETSAT and EUMETCAL, the EUMETNET education and training collaborative network of the European NMHSs. Training programs are designed to enhance the understanding and application of ECMWF advanced weather prediction systems and data, thereby improving weather forecasting capabilities globally, especially for its Member and Co-operating States. Between 2022 and 2024, ECMWF held 27 courses (on-site and online) with more than 400 participants in total every year. The average participation from NMHSs was 55% in 2024. In addition, ECMWF also offers training through Massive Open Online Courses (MOOC),

the latest being on ML in weather and climate in 2023, with more than 9000 registrants. ECMWF also offers fellowship positions aimed at supporting individuals in research and professional development related to weather forecasting, climate research, and related fields.

6. Infrastructure, software tools, applications and services

Underpinning all ECMWF activities are the computing infrastructure and software tools (including HPCF, software applications, cloud etc.) allowing research to be carried out, vast amounts of data to be analysed, forecasting data to be produced and disseminated in real-time and collaborations/partnerships to be sustained. In return, many research activities provide feedback on the computing infrastructure to help improving its capacity or creating new applications for weather forecasting operations.

The HPCF is essential for the operational and research activities of ECMWF. Data assimilation, dynamical equation solving, parametrization, and ensemble components of climate monitoring and weather/atmospheric composition forecasting are computationally intensive processes. The HPCF allows for faster model simulations, increased resolutions (Fig. 4), processing of vast ensembles of climate and weather scenarios, leading to more accurate, reliable and timely predictions. 75% of the capacity of the HPCF is dedicated to ECMWF's activities (25% for the Centre's time-critical operations and 50% for its research activities).

In 2017 ECMWF commissioned a socio-economic benefit internal report to Citizing to evaluate the incremental benefits that would be delivered by different levels of HPCF investment, concluding that the benefits deriving from an option that was considered "intermediate" were estimated at around five billion pounds over the envisaged four-year contract period. This estimate was capturing only the incremental value to products and services enabled by the HPCF upgrade and not the underlying value present before such upgrade and did not attempt to capture the value deriving from the overall ECMWF activities described in this paper.

In October 2022, ECMWF started to use its new HPCF provided via a service contract with Atos and featuring over one million processors spread across 4 clusters and a performance of about 30 petaflops (10¹⁵ or a million billion calculations per second). The increased medium-range ENS resolution from 18 km to 9 km implemented in 2023 thanks to the new HPCF, led to a 2.5 factor increase in the volume of data processed, compared to the previous cycle.

ECMWF makes 25% of its HPC capacity available to its Member States for their time-critical operations and research activities, including for NWP application development and production of software, pre- and post-processing, computational research, and to be used as part of their disaster recovery plans. In 2020, for instance, after a strong earthquake destroyed the headquarters of DHMZ (the Croatian NMHS) in Zagreb, the time-critical operations could be rapidly backed up on ECMWF HPCF and IT infrastructure, allowing the NMHS to continue operating to provide forecasts and other services to the nation.

ECMWF's HPCF is also used to support the WMO. The South-East European Multi-Hazard Early Warning Advisory System (SEE-MHEWS-A), for instance, is an initiative started in 2014 by WMO, designed to prepare for and respond effectively to natural hazards and extreme weather events by improving forecasting and warning provisions across 19 countries in Southeast Europe (WMO, 2024). The platform, implemented using ECMWF's HPCF and the European Weather Cloud (EWC), currently allows forecasters from the countries involved to work collaboratively to identify potential hazards and impacts, especially when impending weather events may affect multiple countries. Since the launch of the SEE-MHEWS-A initiative, data exchange in the region has increased by 50%, and further improvement is expected, including incorporating other types of data, such as hydrological observations and radar data (WMO, 2024).

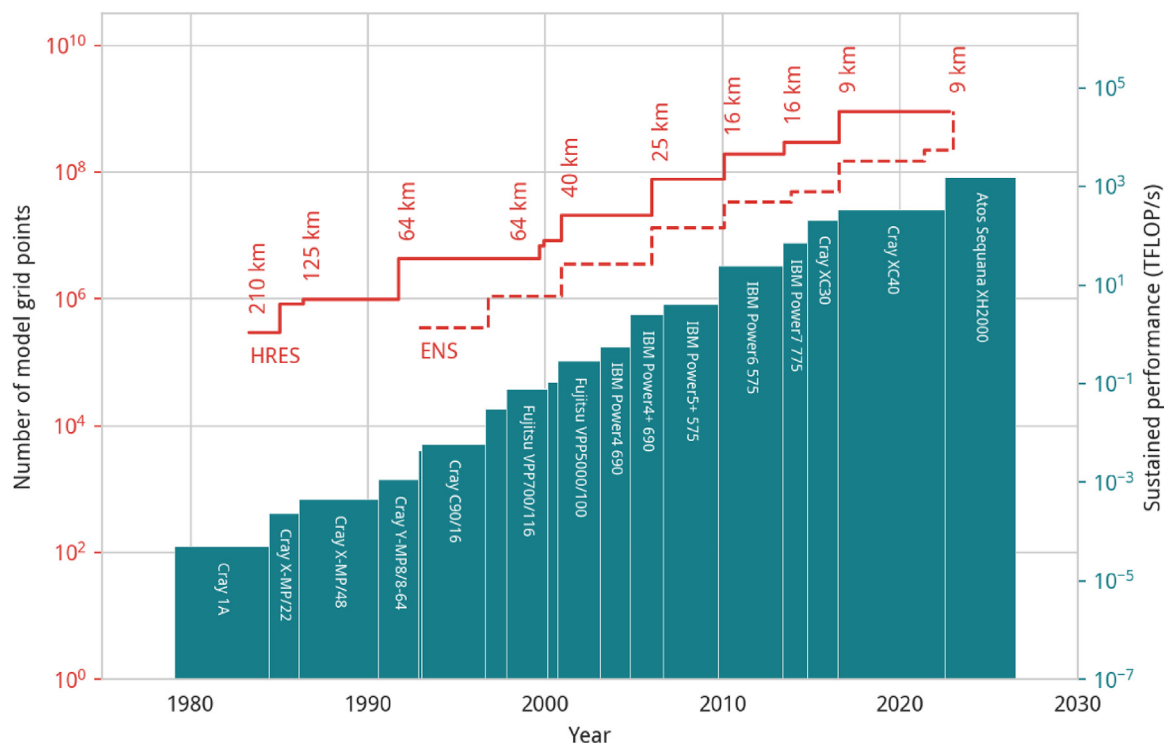


Fig. 4. Evolution of sustained CPU performance of ECMWF's HPC. The advanced research carried out at ECMWF and its application in an operational environment are computationally expensive. The production of high-resolution ensembles of global weather forecasts would not be possible without the investment made by the Member States over ECMWF's lifetime.

Cloud solutions have been increasingly adopted by ECMWF in the last few years, enabling the management of vast amounts of data, as well as running processing software closer to the data, facilitating the development of new and more efficient forecasting services. While the solutions provided to different typologies of users differ, the underlying cloud infrastructure is gradually being harmonized. The aforementioned EWC is mainly dedicated to supporting the NMHSs of the Member and Co-operating States. A joint initiative between ECMWF and EUMETSAT, the EWC is a collaboration platform for meteorological application development and operations in Europe enabling the digital transformation of the European Meteorological Infrastructure (EMI), comprising ECMWF, EUMETSAT, EUMETNET and their respective members to support them in fulfilling their official duties and related R&D activities. The EWC can also be linked to public clouds to allow further data accessibility and flexibility. For instance, the EWC is used to host one of the core components of the framework for data sharing utilized by WMO, called WIS2.0, during its pre-operational phase. This shows the ability of the EWC to interconnect with other clouds and services and demonstrates its more outward-facing aspects compared to the HPCF (EWC, 2024).

The EWC and other cloud-based systems such as the Copernicus funded Climate Data Store (CDS) and Atmosphere Data Store (ADS) are based on a shared platform, the Common Cloud Infrastructure. CDS and ADS host and disseminate the outputs of C3S, CAMS and, in part, CEMS. The infrastructure also provides toolkits to execute some post-processing and Jupyter Notebooks to explore the datasets available and for training purposes.

Destination Earth (DestinE) is an initiative to digitally enable European environmental programmes and aims to develop a highly accurate digital replica of the Earth, called digital twin, to model and simulate natural phenomena, hazards, and related human activities influencing them. The initiative is implemented by ECMWF, ESA and EUMETSAT, and with the participation of many European organisations as contractors. DestinE attempts to unlock the potential of digital modelling of the Earth system at a level that represents a real breakthrough

in terms of accuracy, local detail, access-to-information speed and interactivity. Using an unprecedented amount of data, innovative Earth system models and cutting-edge computing, DestinE will allow users to explore interactively the different components of the Earth system (land, marine, atmosphere, biosphere) and natural and human-induced change, looking at the past and present and test and develop future scenarios. ECMWF is responsible for implementing two high-priority digital twins (on climate adaptation and weather extremes) and the Digital Twin Engine (DTE), the software and data environment needed to power the digital twins. The initiative is also investing in dedicated technology partnerships with HPC centres, vendors and international organisations representing the entire technology range across extreme computing, big data, smart sensors and networks. DestinE ties into EuroHPC, a € 1 billion project to build a series of pre-exascale and exascale supercomputers across Europe, including Finland's LUMI, Italy's LEONARDO, Spain's MARENOSTRUM 5, Luxembourg's MELUXINA, Czechia's KAROLINA, Bulgaria's DISCOVERER, Slovenia's VEGA, Portugal's DEUCALION and Germany's JUPITER. EuroHPC is tasked to run the digital twins and the DTE and in June 2024, the European Commission launched DestinE at CSC in Finland, home of LUMI.

The increasing complexity of Earth-system prediction models, the ever-growing volume of data required, and evolving computing technologies need a constant upgrade of hardware and software architectures. For instance, the ongoing, overarching trend is the shift from purely CPU-based platforms to heterogeneous systems with GPU accelerators. Studies suggest that GPUs provide substantial speedups and efficiency, such as nearly 193 times faster computations for intensive portions of models like NOAA's Weather Research and Forecasting (WRF) model using NVIDIA GPUs (Huang et al., 2015). Additionally, the adoption of GPUs for weather prediction allows for forecasts to be issued two times faster compared to CPU-based models, showcasing the computational power of GPUs (Batista and Pinzon, 2023; Xiao et al., 2022). Furthermore, the feasibility and benefits of GPU acceleration for NWP codes have been explored, indicating that GPU and multicore acceleration can be both feasible and advantageous for enhancing

weather modelling capabilities (Vanderbauwhede and Takemi, 2015). The very first efforts to run NWP models on GPUs were made by MeteoSwiss in the framework of the COSMO Consortium (COSMO, 2024). To address the transition from CPUs to GPU, ECMWF has been exploring the use of GPUs to run IFS more efficiently. ECMWF's HPCF utilizes NVIDIA's next-generation GPU architecture and is supported by Atos Codex AI Suite, the most comprehensive AI software suite on the market. This provides advanced GPU computing and AI capability, enabling ECMWF researchers to speed up processing times on the most complex data, allowing them to gain faster insights and reduce forecasting production costs using the power of deep learning and analytics. The Centre of Excellence in HPC, AI and Quantum Computing for Weather & Climate, in partnership with Atos, allowed ECMWF to start the development of a hybrid CPU/GPU-based version of IFS and to prepare ECMWF's product-generation pipeline and data-centric workflows for new technologies. ECMWF is currently working on a key component of IFS called FVM (intended to provide modelling flexibility at higher resolutions) by running it on multiple GPUs. More specifically, ECMWF is currently collaborating with Swiss partners, rewriting and further developing FVM with the domain-specific library GT4Py of the Swiss National Supercomputing Centre (CSCS) and ETH Zurich (ECMWF, 2023a). Preliminary results indicate the potential of the GT4Py approach to enhance portability and productivity on the GPU adaptation of the IFS.

Beyond data access and computing services (which include hardware, management and support), ECMWF offers specialized software packages tailored to the needs of NMHSs. ECMWF's robust software suite bolsters NMHSs' weather forecasting capabilities and scientific research and is now collectively known as ECMWF Software Engine (ESEE). This comprehensive toolkit offers functionalities across various aspects of weather analysis and prediction. ECMWF Member States have access to a powerful suite of software tools for efficiently retrieving, processing, and analysing the vast meteorological data archive maintained by ECMWF. This empowers them to not only leverage ECMWF's data for their own forecasting models but also seamlessly integrate them with their national observational networks and research datasets. Tools like ECaccess provide a single point of access to ECMWF's computing and archiving facilities. Member States can retrieve vast datasets seamlessly through ECaccess for further analysis. MARS (Meteorological Archival and Retrieval System) empowers efficient access, processing, and analysis of ECMWF's archived meteorological data. Currently, there are over 7,000 MARS registered users. Licences have been granted to projects in about 40 countries worldwide. ECMWF has recently observed a significant growth in the number of projects relating to artificial intelligence and machine learning but also, for instance, to the use of ensemble forecasts for hazard forecasting and long-term forecasts for energy prediction. Other tools may encompass software for data assimilation or post-processing forecast results. Metview, for instance, allows NMHSs to tailor the information for specific regional applications or user needs, as well as visualize and manipulate the data for effective communication to stakeholders.

Recognizing the broader scientific research needs of its members, ECMWF maintains a comprehensive suite of software tools for a variety of tasks. This software suite, co-developed with Member States, allows them to leverage ECMWF's expertise in scientific computing for research endeavours beyond just weather forecasting.

For example, ECMWF relies heavily on its Scalable Acquisition and Pre-Processing (SAPP) system as it is a core component of its operational workflow for weather forecasting. SAPP acts as the central hub for the acquisition of observational data from various sources and different formats. These include meteorological observations from ground stations, ships, aircraft, and satellites. While SAPP is primarily used by ECMWF for its own operational forecasting, since 2018, ECMWF has opened SAPP to its Member States through an optional programme (ECMWF, 2019) allowing NMHSs to leverage the capabilities of SAPP for their forecasting systems and therefore potentially

significantly reduce the cost of developing and maintaining their own data acquisition infrastructures. The user community collaborates in the development of new decoders designed to translate data from new observation types and formats into the standardized BUFR format.

ECMWF is also engaging in projects with the NMHS of its Member and Co-operating States to develop software toolkits to acquire and process Internet of Things (IoT) observations, to adapt ECMWF software to emerging technologies and enable its adoption by other organisations, to develop the previously mentioned Anemoui platform for ML models, and to dissemination data platform via the recently released openECPDS.

7. Conclusions and new ECMWF strategy

More than ever, society is dependent on accurate weather and climate data and the ability of research institutions, governmental agencies, commercial companies and other players to integrate them into their own processes to provide actionable information to influence decisions that can have a positive, profound and long-lasting societal impact. ECMWF, with its products, infrastructure and services and the strong, deep and far-reaching links with its Member States and, more generally, the meteorological and environmental sciences ecosystem, is uniquely positioned to provide a fundamental contribution to addressing the environmental challenges ahead. The evolution of the landscape, as mentioned in section 1, however, required a new long-term strategy, which was developed in 2024 to cover the period 2025–2034 (ECMWF, 2025). It is paramount that ECMWF remains a leader in global NWP, prioritizing high-impact weather forecasting and integration of ML in scientific developments. ECMWF needs to expand activities where benefits for the efficient operation of the European partners can be realized, innovating at the cutting edge of physical, computational and data science for environmental monitoring and prediction and exploiting data-driven methods anchored on physics-based modelling. Until 2022 ECMWF, like many other traditional players in the meteorological field, considered that the main disruptive impact of AI/ML would be to accelerate or improve components of the existing NWP setup (e.g. in data assimilation, observation operators, data monitoring etc.). ECMWF has since embraced AI/ML techniques also as an alternative to physics-based simulations, in particular but not exclusively for the forecast itself, with the first operational implementation of its AIFS model in February 2025. This first operational implementation of an ML model in our community paves the way to future applications of AI/ML for ensemble forecasting at all ranges and extensions to the Earth system, as well as developments in predicting the weather directly from observations and the development of a foundation model.

The difficulty and complexity of accessing large amounts of data are often highlighted by users in their interactions with ECMWF. While network, hardware and software updates are regularly implemented to address these issues, their benefits are offset by the increase in the availability of new and improved datasets, ever-increasing resolutions and ever-expanding and demanding user base. ECMWF will, therefore, continue its efforts in developing cloud services and maximization of the benefits deriving from proximity to data, portability, interoperability, and federation. Greater emphasis will be put on collaborations, primarily with the European Meteorological Infrastructure, through existing and new mechanisms, particularly in new and challenging scientific and technological domains, by co-developing software tooling and applications that can be exploited by the wider community and at the same time contribute to building expertise. In the spirit of greater collaboration and to break the barriers to a broader uptake of its products, in the new strategy, ECMWF is accelerating the adoption of open policies for data and software, recognizing that many other players in the meteorological and climatological sectors have already demonstrated some of the substantial benefits that can be derived.

CRediT authorship contribution statement

Fabio Venuti: Writing – review & editing, Writing – original draft, Project administration, Conceptualization. **Florence Rabier:** Writing – review & editing, Supervision. **Erik Andersson:** Writing – review & editing, Project administration. **Umberto Modigliani:** Writing – review & editing. **Stephen English:** Writing – review & editing. **Christine Kitchen:** Writing – review & editing. **Matthieu Berrone:** Writing – original draft. **Dorian Pinsault:** Project administration. **Julie Debrux:** Conceptualization. **Michel Jarraud:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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